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ACADEMIC BACKGROUND

DEGREE	FIELD of Study	Institution	Country	Date
B.Tech (Bachelor of Technology)	Computer Science & Engineering	K. K. College of Engineering & Management, Dhanbad	India	2023 – 2027

Relevant coursework: Data Structures & Algorithms, Artificial Intelligence & Machine Learning, Information Retrieval, Data Mining, Operating Systems, Computer Networks, Databases, Distributed Systems, Computer Organisation, Discrete Mathematics.

ACADEMIC, RESEARCH and INDUSTRIAL EXPERIENCE

Position	Organization	Department	Period
Research & Development Intern	Indian Institute of Technology Kharagpur	Department of Artificial Intelligence Supervisor: Dr. Koustav Rudra	May 2026 – Jun 2026
Research & Development Project Intern	Indian Institute of Technology (ISM) Dhanbad	Computer Science & Engineering Supervisor: Dr. Sachin Tripathi	Feb 2026 – Apr 2026
Deep Learning Intern	Birsa Institute of Technology, Sindri	Supervisor: Dr. Khustar Ansari	Jun 2025 – Jul 2025
Software Engineer (Part-time)	VaporFi Labs (Remote)	Engineering	Sep 2022 – Apr 2026

DESCRIPTION OF ACTIVITIES AT THE ACADEMIC INSTITUTIONS

Published Papers

- "When Better Eyes Lead to Blindness: A Diagnostic Study of the Information Bottleneck in CNN-LSTM Image Captioning Models," *International Journal of Computer Applications*, 2025. DOI: 10.5120/ijca2025925560. Cited 3 times, including in IEEE-published research, IOP Science and the NCIT Journal.

Research Projects

- "Synthetic QA & MCQ Dataset Construction from YouTube Cryptocurrency Video Transcripts," Prof. Koustav Rudra, Department of Artificial Intelligence, IIT Kharagpur, 2026. Built the first question-answering and multiple-choice dataset derived from long-form spoken cryptocurrency video transcripts, released as the official data for the CryptOQA shared task at FIRE 2026. Designed the end-to-end construction pipeline over a corpus of 1,145 ASR transcripts: a seven-pass cleaning stage, controlled chunking experiments (recursive sentence-aware versus semantic chunking), dual-route question generation with Qwen2.5-14B-Instruct served through vLLM, LLM-as-judge faithfulness filtering using a deliberately cross-family judge (Gemma-3-27B) to avoid self-preference bias, embedding-based de-duplication, and a leakage-safe, topic- and hop-stratified split meeting exact target sizes with verified zero cross-split leakage. Cut a full generation pass from over 24 hours to roughly three by migrating from HuggingFace static batching to vLLM's paged KV-cache and continuous batching. Built a two-stage MCQ ambiguity detector intersecting an NLI paraphrase signal with oracle-model disagreement, measuring a 3.7% defect rate. Ran a meta-evaluation of lexical, semantic and learned-equivalence metrics against human correctness labels, establishing NLI-based answer equivalence as the only measure that separates correct paraphrases from same-

vocabulary contradictions (AUROC 0.80 on real predictions, versus 0.52–0.74 for the alternatives), and recommended it as the shared task's primary scoring metric. Final release: 3,500 items across both sub-tasks.

- **"Hyperledger Fabric, Blockchain-Assisted Healthcare EHR and Fog Computing,"** Prof. Sachin Tripathi, Department of Computer Science & Engineering, IIT (ISM) Dhanbad, 2026. Carried out under MeitY-funded project MeitY/2024-25/1106/CSE. Surveyed the 2018–2026 literature on blockchain-based electronic health record systems and identified three gaps unaddressed by any single existing system: on-chain verification of medical crowdfunding flows, consent handling for emergency access when a patient cannot consent, and the absence of HL7 FHIR compliance in existing prototypes. Built a proof-of-concept EHR system on a two-organisation Hyperledger Fabric network integrating a client-side blockchain oracle, Node.js chaincode emitting events on a pending lab-report request, a Fabric Gateway listener ingesting off-chain data and submitting signed fulfilment transactions, and a Next.js interface with role-separated views for doctor, patient, oracle monitor, and network explorer — demonstrating a complete pending-to-completed state transition driven by external data. Separately implemented the BAloG protocol (Tomar & Tripathi, *Cluster Computing*, 2022) for mutual authentication and key agreement in a three-layer smart-grid fog architecture: a single-organisation Fabric network with two channels enforcing identity anonymity, chaincode covering all six registration transaction functions, ECC-based pseudo-identity and credential generation, and the four-message handshake yielding a shared session key across smart meter, fog node, and cloud server, verified at the application level.
- **"Deployable Deep Learning Pipelines for Image Processing,"** Dr. Khustar Ansari, Birsa Institute of Technology, Sindri, 2025. Translated academic research papers into production-ready Python software artifacts; led a team of four in defining model-training and cloud-deployment standards, achieving a 100% deployment success rate across all systems.

Other Relevant Activities

- **Track Coordinator**, shared task on cryptocurrency information retrieval, organised by IIT Kharagpur and IISER Kolkata at the Forum for Information Retrieval Evaluation (FIRE) 2026.
- **arXiv endorsement (cs.CV, cs.LG)** — endorsed by established computer scientists for preprint publication, obtained independently without institutional affiliation.
- **First Rank**, Computer Science & Engineering programme, K. K. College of Engineering & Management.
- **Hindustan Talent Honour 2023** — recognised for outstanding academic excellence.
- **Technical proficiencies:** Python, TypeScript/JavaScript, SQL, Solidity, Bash; PyTorch, TensorFlow, Transformers, vLLM, Hugging Face; Node.js, FastAPI, GraphQL, REST, PostgreSQL, Redis; Docker, Kubernetes, CI/CD, Linux.

AREAS OF EXPERTISE

Natural language processing, large language models, information retrieval, deep learning.